INTERIOR $C^{1,\beta}$ ESTIMATES FOR THE INHOMOGENEOUS INFINITY-LAPLACE EQUATION

ABSTRACT. We prove quantitative interior gradient estimates for viscosity solutions of the unnormalized inhomogeneous infinity-Laplace equation $-\Delta_\infty u = f$ in arbitrary dimension $n \geq 2$. Let $a = a_{n+1} \in (0, 1/3]$ be an exponent in the homogeneous interior gradient estimate of OpenAI in dimension $n+1$, which is used as an established theorem. If $f \in C^{0,\sigma}$, $0 < \sigma \leq 1$, then $u \in C^{1,\beta}_{\mathrm{loc}}$ with $\beta = a(1+\sigma)/(2+\sigma+a)$, and we obtain a scale-invariant estimate depending only on the oscillation of u and the cubic-root size of the source. Continuous bounded sources already give the exponent $a/(2+a)$, while constant sources retain the homogeneous exponent a. In particular, $f \in C^{0,\alpha}$ implies $u \in C^{1,\alpha}_{\mathrm{loc}}$ whenever $0 < \alpha \leq \sqrt{1+a} - 1$. The proof uses smooth one-dimensional profiles to absorb a frozen source in an additional variable. Homogeneous comparison gives a linear error bound in the variation of that source, which is combined with the homogeneous gradient estimate and a direct affine-approximation argument. All sources may change sign, and no nonvanishing-gradient assumption is made. An explicit constant-source solution shows that exponents above $1/3$ are impossible even when the gradient is bounded away from zero.

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1. Introduction and main results

We study bounded continuous viscosity solutions of

$$-\Delta_\infty u = f \qquad \text{in } \Omega \subset \mathbb{R}^n, \qquad n \geq 2, \tag{1.1}$$

where throughout this paper

$$\Delta_\infty u = \langle D^2 u\, Du, Du\rangle = \sum_{i,j=1}^{n} u_i u_j u_{ij} \tag{1.2}$$

is the unnormalized infinity-Laplacian. The source f is allowed to vanish and to change sign. The coefficient matrix in (1.2) is $Du \otimes Du$, so the equation controls only the second derivative in the gradient direction. Even a positive lower bound for $|Du|$ does not make the equation uniformly elliptic. Our objective is a quantitative Hölder estimate for the full vector gradient.

The normalization of the operator is essential. The equation with $|Du|^{-2}\Delta_\infty u$ and a prescribed nonzero source is a different equation, with different scaling and a separate convention at zero gradient. All statements below concern (1.2).

1.1. **Research context.** The homogeneous infinity-Laplace equation originates in the theory of absolutely minimizing Lipschitz extensions. Jensen established the fundamental uniqueness theory [8]; a broad account of the variational and viscosity formulations is given by Aronsson, Crandall, and Juutinen [1]. An elementary proof of the homogeneous comparison principle was subsequently given by Armstrong and Smart [2].

For homogeneous regularity, Savin proved local C^1 regularity in the plane [14], and Evans and Savin obtained a positive gradient Hölder exponent there [5]. Evans and Smart proved everywhere differentiability in arbitrary dimensions [6]. Recent planar work addresses the exponent $1/3$. Brustad obtained that exponent under nonvanishing of the gradient and the quantitative injectivity condition $|Du(x) - Du(y)| \geq c|x - y|$ with $c > 0$ [3]. Xu's July 2026 preprint proves qualitative $C^{1,1/3}_{\mathrm{loc}}$ regularity for general planar infinity-harmonic functions [16, Theorem 1.1]. Sobral, Teixeira, and Urbano then obtain uniform planar interior estimates for every exponent strictly below $1/3$ [15, Theorem 1.3]. Their quantitative statement does not assert a uniform estimate at the endpoint. The literature discussion here includes sources available on October 17, 2026.

For the inhomogeneous equation, Lu and Wang developed existence, comparison, and stability results for sources bounded away from zero, as well as stability of the homogeneous equation under perturbations [12]. Lindgren showed that first-order blow-up limits are affine for bounded continuous sources and established everywhere differentiability when the source is C^1 [10, Theorems 1 and 2]. These statements concern affine blow-ups and differentiability, and do not include a uniform Hölder estimate for the vector gradient. In the plane, Koch, Zhang, and Zhou proved sharp Sobolev estimates for powers of $|Du|$ when $f \in BV_{\mathrm{loc}} \cap C^0$ and $|f| > 0$ [9]; in particular, $|Du|^q \in W^{1,2}_{\mathrm{loc}}$ for $q > 3/2$. For the special problem with source 1 and zero boundary data in bounded convex domains satisfying an interior sphere condition, Crasta and Fragalà established power concavity and C^1 regularity [4].

Separated-variable lifting is already present in this literature. Hong and Feng proved a superposition principle when one summand is smooth [7], and Liu, Manfredi, and Zhou proved the general disjoint-variable superposition property for viscosity solutions [11]. We use the smooth-summand case and include its short viscosity proof. The quantitative step is to vary the source parameter of the smooth profile, construct homogeneous subsolutions and supersolutions, and control the difference from a homogeneous replacement.

The homogeneous theorem of OpenAI [13] provides the higher-dimensional regularity estimate needed for this construction. We use that result as an established theorem, with exactly the dependence on dimension and oscillation stated below. The purpose of this paper is the quantitative

transfer to (1.1). The earlier results just described provide context and attribution; the proof does not require an inhomogeneous differentiability or gradient continuity theorem.

1.2. **The homogeneous input and the main estimate.** For a set E and $0 < \gamma \le 1$, write

$$[g]_{\gamma;E} = \sup_{\substack{x,y\in E\\ x\neq y}} \frac{|g(x)-g(y)|}{|x-y|^\gamma}, \qquad \operatorname{osc}_E u = \sup_E u - \inf_E u.$$

The notation $B_r^m(z)$ specifies the ambient dimension when needed; we omit the superscript when the dimension is clear. Here $C^{0,\gamma}(E)$ denotes the space of bounded continuous functions with finite seminorm $[g]_{\gamma;E}$, while $C^{0,\gamma}_{\mathrm{loc}}(\Omega)$ requires these bounds only on compact subsets of Ω. In particular, the data norms used in the normalizations below are finite.

Theorem 1.1 (OpenAI, homogeneous interior estimate)**.** *For each integer $d \ge 3$, there are constants $a_d \in (0, 1/3]$ and $C_d < \infty$, depending only on d, such that every bounded infinity-harmonic function $h \in C(B_1^d)$ belongs to $C^{1,a_d}(B_{1/2}^d)$ and satisfies*

$$\|Dh\|_{L^\infty(B^d_{1/2})} + [Dh]_{a_d;B^d_{1/2}} \le C_d \operatorname{osc}_{B_1^d} h. \tag{1.3}$$

This is [13, Theorem 1.1]. The exponent a_d is not explicit. For the rest of the paper, fix $n \ge 2$ and set

$$a = a_{n+1}, \qquad C_H = C_{n+1}. \tag{1.4}$$

The use of dimension $n+1$ comes from the auxiliary variable in the proof.

Theorem 1.2 (Hölder sources)**.** *Let $0 < \sigma \le 1$, and let $u \in C(B_R(x_*))$ be a bounded viscosity solution of (1.1) in $B_R(x_*) \subset \mathbb{R}^n$, where $f \in C^{0,\sigma}(B_R(x_*))$. Set*

$$\beta = \frac{a(1+\sigma)}{2+\sigma+a}, \qquad F_R = \|f\|_{L^\infty(B_R(x_*))}, \qquad H_{R,\sigma} = [f]_{\sigma;B_R(x_*)}. \tag{1.5}$$

Then $u \in C^{1,\beta}(B_{R/2}(x_*))$, and

$$\begin{aligned} R\|Du\|_{L^\infty(B_{R/2}(x_*))} + R^{1+\beta}[Du]_{\beta;B_{R/2}(x_*)} \\ \le C\left\{ \operatorname{osc}_{B_R(x_*)} u + \left(R^4 F_R + R^{4+\sigma} H_{R,\sigma}\right)^{1/3} \right\}. \end{aligned} \tag{1.6}$$

Here C depends only on n, σ and the fixed constants a, C_H in (1.4), hence only on n, σ after Theorem 1.1 is fixed.

The theorem applies on every ball compactly contained in an open domain, and therefore yields the corresponding local statement. No sign or nonvanishing condition is imposed on the source or the gradient. The cubic-root dependence on f is consistent with $\Delta_\infty(Au) = A^3\Delta_\infty u$; the spatial factor R^4 follows from $\Delta_\infty(u(x_* + R\,\cdot)) = R^4(\Delta_\infty u)(x_* + R\,\cdot)$.

We also prove three useful consequences in Section 5. Continuous bounded sources give the positive exponent

$$\beta_0 = \frac{a}{2+a} \tag{1.7}$$

with an estimate involving only $\operatorname{osc} u$ and $\|f\|_\infty$. For constant sources, an exact lift retains exponent a. Finally, if

$$0 < \alpha \le \alpha_* := \sqrt{1+a} - 1, \tag{1.8}$$

then $f \in C^{0,\alpha}$ implies $u \in C^{1,\alpha}_{\mathrm{loc}}$. The exponent in (1.5) is an explicit function of the homogeneous exponent, but it is not asserted to be optimal. The example in Proposition 5.5 excludes any general exponent above $1/3$, even for a smooth constant source and a gradient bounded away from zero.

1.3. **Outline of the proof.** Section 2 fixes the viscosity convention and proves an interior Lipschitz estimate using a radial barrier. At scale r, this estimate controls the first-order rescaling $u_r(y) = (u(x_0 + ry) - u(x_0))/r$, whose source is $rf(x_0 + ry)$. Section 3 introduces smooth profiles W_c with $(W_c')^2 W_c'' = c$ and proves that $u_r + W_c$ has source $rf(x_0+ry)-c$. Choosing $c = rf(x_0)$ and varying it by the source oscillation produces a homogeneous replacement whose error, in the original variables, is $O(r^{2+\sigma})$.

Theorem 1.1 gives a Taylor error $O(\rho^{1+a} r^{-a})$ on a smaller ball of radius ρ. Section 4 balances these errors by taking $r = \rho^{(1+a)/(2+\sigma+a)}$. The resulting affine error is $O(\rho^{1+\beta})$. A direct comparison of successive affine slopes proves differentiability at every point, and comparison of the resulting Taylor expansions proves the uniform Hölder estimate for Du. Section 5 proves the companion results and the 1/3 obstruction.

2. Preliminaries and a local Lipschitz estimate

2.1. Viscosity convention and homogeneous comparison.

Definition 2.1. Let $f \in C(\Omega)$. A continuous function u is a viscosity subsolution of $-\Delta_\infty u = f$ if every C^2 function ϕ touching u from above at $x_0 \in \Omega$ satisfies

$$-\langle D^2\phi(x_0) D\phi(x_0), D\phi(x_0)\rangle \le f(x_0).$$

For a supersolution, lower tests give the reverse inequality. A viscosity solution is both a subsolution and a supersolution. Tests with $D\phi(x_0) = 0$ are included.

In particular, an upper test with zero gradient forces $0 \le f(x_0)$, and a lower test with zero gradient forces $0 \ge f(x_0)$.

Lemma 2.2 (Viscosity scaling)**.** *Suppose $u \in C(\Omega)$ solves $-\Delta_\infty u = f$ with $f \in C(\Omega)$. Let $x_0 \in \mathbb{R}^n$, $r, A > 0$, $b \in \mathbb{R}$, and $\Omega_{x_0,r} = \{y : x_0 + ry \in \Omega\}$. Then*

$$v(y) = \frac{u(x_0 + ry) - b}{A} \quad \Longrightarrow \quad -\Delta_\infty v(y) = \frac{r^4}{A^3} f(x_0 + ry) \quad \text{in } \Omega_{x_0,r} \tag{2.1}$$

in the viscosity sense. Also, $-u$ solves $-\Delta_\infty(-u) = -f$.

Proof. If ψ is an upper test for v at y_0, then

$$\phi(x) = b + A\psi\big((x - x_0)/r\big)$$

is an upper test for u at $x_0 + ry_0$. At the corresponding points, $D\phi = (A/r)D\psi$ and $D^2\phi = (A/r^2)D^2\psi$, so

$$-\Delta_\infty \psi(y_0) = \frac{r^4}{A^3}(-\Delta_\infty \phi)(x_0 + ry_0) \le \frac{r^4}{A^3} f(x_0 + ry_0).$$

Lower tests give the reverse inequality, proving (2.1). For the last assertion, an upper test ψ for $-u$ corresponds to the lower test $-\psi$ for u. Since $\Delta_\infty(-\psi) = -\Delta_\infty \psi$, the supersolution inequality for u becomes the subsolution inequality for $-u$ with source $-f$. The other inequality follows in the same way. □

We use the following standard homogeneous fact. For continuous boundary data b on a Euclidean ball B, the Dirichlet problem

$$-\Delta_\infty h = 0 \text{ in } B, \qquad h = b \text{ on } \partial B \tag{2.2}$$

has a unique solution $h \in C(\overline{B})$. Moreover, if $S, T \in C(\overline{B})$ are respectively a subsolution and a supersolution of the homogeneous equation and $S \le b \le T$ on ∂B, then

$$S \le h \le T \qquad \text{in } B. \tag{2.3}$$

These are standard consequences of the homogeneous Dirichlet theory [8, 1]; see also [2] for the comparison principle. Comparison with constants gives $\operatorname{osc}_B h \le \operatorname{osc}_{\partial B} b$. Every use of a Dirichlet comparison in this paper is of the form (2.3).

2.2. **A radial barrier.**

Lemma 2.3 (Interior Lipschitz bound)**.** *Suppose $u \in C(\Omega)$ solves $-\Delta_\infty u = f$ with $f \in C(\Omega)$. Let $\overline{B_d(x_0)} \subset \Omega$, and set*

$$M = \operatorname{osc}_{B_d(x_0)} u, \qquad F = \|f\|_{L^\infty(B_d(x_0))}.$$

Then, for $x \in B_d(x_0)$,

$$|u(x) - u(x_0)| \le \Big[(M/d)^3 + 3Fd\Big]^{1/3} |x - x_0| \le \left(\frac{M}{d} + (3Fd)^{1/3}\right) |x - x_0|. \tag{2.4}$$

In particular, if u solves the equation in B_1 and satisfies

$$\operatorname{osc}_{B_1} u \le 1, \qquad \|f\|_{L^\infty(B_1)} \le 1, \tag{2.5}$$

then

$$\operatorname{Lip}(u; B_{3/4}) \le L \tag{2.6}$$

for a numerical constant L; one may take $L = 9$.

Proof. Fix $\varepsilon > 0$ and define, for $0 \le s \le d$,

$$q(s) = \int_0^s \Big[(M/d)^3 + 3(F + \varepsilon)(d - \tau)\Big]^{1/3} \, d\tau. \tag{2.7}$$

The integrand is continuous on $[0, d]$ and bounded below by M/d. Thus $q \in C^1([0, d])$, $q(0) = 0$, and $q(d) \ge M$. The radicand is strictly positive for $0 \le s < d$, so q is smooth there and

$$q'(s) > 0 \quad (0 \le s < d), \qquad -(q'(s))^2 q''(s) = F + \varepsilon \quad (0 < s < d).$$

For the radial function $Q(x) = u(x_0) + q(|x - x_0|)$, the gradient is radial; therefore its infinity-Laplacian contains only the radial second derivative:

$$-\Delta_\infty Q = -(q')^2 q'' = F + \varepsilon \qquad \text{in } B_d(x_0) \setminus \{x_0\}.$$

Continuity on the closed ball implies that its oscillation agrees with the oscillation on the open ball. Hence $u(x) - u(x_0) \le M \le q(d)$ on $\partial B_d(x_0)$, so $Q \ge u$ there, while $Q(x_0) = u(x_0)$. If $u - Q$ had a positive maximum on the closed ball, a maximizing point would lie in the punctured open ball. There Q plus a constant would be a smooth upper test for u, contradicting $F + \varepsilon \le f \le F$. It follows that $u(x) - u(x_0) \le q(|x - x_0|)$.

Since q' is nonincreasing, $q(s) \le q'(0)s$. By Lemma 2.2, the same construction applies to $-u$ with source $-f$. Letting $\varepsilon \downarrow 0$ yields both one-sided bounds. This proves the first inequality in (2.4); the second follows from $(A + B)^{1/3} \le A^{1/3} + B^{1/3}$ for $A, B \ge 0$.

Under (2.5), take $d = 1/8$ at any center in $B_{3/4}$. The closed comparison ball lies in B_1, and the first coefficient in (2.4) is at most $(8^3 + 3/8)^{1/3} < 9$. For arbitrary $x, y \in B_{3/4}$, subdivide their line segment into finitely many pieces of length strictly less than $1/8$. Every subdivision point belongs to the convex ball $B_{3/4}$, and the preceding centered estimate applies to each consecutive pair. Summing proves (2.6). □

3. Smooth lifting and homogeneous approximation

3.1. **The exact viscosity identity.** The next lemma is the smooth-summand case of the superposition principle [7, 11]. We give a direct proof, including tests with zero horizontal gradient.

Lemma 3.1 (Smooth separated-variable lifting)**.** *Let $m \geq 1$, let $G \subset \mathbb{R}^m$ be open, let $I \subset \mathbb{R}$ be an open interval, and let $v \in C(G)$ solve $-\Delta_\infty v = g$ in G, with $g \in C(G)$. For $w \in C^2(I)$, define $V(y,t) = v(y) + w(t)$. Then V is a viscosity solution in $G \times I$ of*

$$-\Delta_\infty V = g(y) - (w'(t))^2 w''(t). \tag{3.1}$$

Proof. Let ϕ touch V from above at (y_0, t_0), with equality at the contact point. Set

$$p = D_y\phi(y_0, t_0), \qquad k = w'(t_0).$$

The vertical slice of the contact gives $\phi_t(y_0, t_0) = k$. Suppose first that $p \neq 0$, and set $b = kp/|p|^2$. On a sufficiently small neighborhood of y_0, the function

$$\psi(y) = \phi(y, t_0 + b \cdot (y - y_0)) - w(t_0 + b \cdot (y - y_0))$$

is a C^2 upper test for v. A large value of b only reduces the size of this neighborhood and causes no restriction on a local viscosity test. For completeness, write the Hessian blocks at the contact point as $H = D^2_{yy}\phi$, $q = D_y\phi_t$, and $\eta = \phi_{tt}$. Since $\phi_t = w' = k$, the chain rule gives

$$D\psi = p, \qquad D^2\psi = H + q \otimes b + b \otimes q + (\eta - w''(t_0))\, b \otimes b.$$

Using $b \cdot p = k$, we therefore obtain

$$\begin{aligned} p^T D^2\psi\, p &= p^T H p + 2k\, q \cdot p + k^2\eta - k^2 w''(t_0) \\ &= (p,k)^T D^2\phi\, (p,k) - k^2 w''(t_0). \end{aligned} \tag{3.2}$$

Thus all mixed horizontal–vertical terms are included. The upper-test inequality for v and (3.2) imply

$$-(p,k)^T D^2\phi\, (p,k) \leq g(y_0) - k^2 w''(t_0).$$

This is the required upper-test inequality for V.

If $p = 0$, the function $y \mapsto \phi(y, t_0) - w(t_0)$ is an upper test for v with zero gradient. Thus $0 \leq g(y_0)$. The vertical contact also gives $\phi_{tt}(y_0, t_0) \geq w''(t_0)$. Consequently,

$$-\Delta_\infty \phi(y_0, t_0) = -k^2\phi_{tt}(y_0, t_0) \leq -k^2 w''(t_0) \leq g(y_0) - k^2 w''(t_0).$$

This proves the subsolution assertion, including $k = 0$. Apply this upper-test argument to $-v$ and $-w$. By Lemma 2.2, $-v$ has source $-g$, so $-V$ is a subsolution with source

$$-g(y) + (w'(t))^2 w''(t) = -(g(y) - (w'(t))^2 w''(t)).$$

An upper test for $-V$ is the negative of a lower test for V; changing signs gives the supersolution inequality for V. This proves (3.1). □

3.2. **Profiles with a source parameter.** For $|d| \leq 2$ and $|t| \leq 1$, define

$$W_d(t) = \int_0^t (8 + 3ds)^{1/3}\, ds. \tag{3.3}$$

The radicand is between 2 and 14, so the profiles are jointly smooth in (d, t) on a neighborhood of $[-2, 2] \times [-1, 1]$. The fixed number 8 ensures that the vertical derivative stays strictly positive uniformly for the parameters used below.

Lemma 3.2 (Profile identities and parameter bounds)**.** *The functions in* (3.3) *satisfy*

$$W_d(0) = 0, \qquad W_d'(t) = (8 + 3dt)^{1/3}, \qquad (W_d')^2 W_d'' = d. \tag{3.4}$$

Moreover,

$$|W_d(t)| \le 14^{1/3}|t|, \qquad 0 \le \partial_d W_d(t) \le 2^{-5/3}t^2 \le 1. \tag{3.5}$$

Consequently, if $\delta \ge 0$ *and* $|c| + \delta \le 2$*, then*

$$0 \le W_{c+\delta}(t) - W_c(t) \le \delta, \qquad 0 \le W_c(t) - W_{c-\delta}(t) \le \delta \quad (|t| \le 1). \tag{3.6}$$

Proof. Differentiation gives (3.4) and the first bound in (3.5). Differentiating with respect to d and substituting $s = t\tau$ gives the identity

$$\partial_d W_d(t) = \int_0^t s(8 + 3ds)^{-2/3}\, ds = t^2 \int_0^1 \tau(8 + 3dt\tau)^{-2/3}\, d\tau.$$

This formula is nonnegative for both signs of t and is at most $2^{-2/3}t^2 \int_0^1 \tau\, d\tau = 2^{-5/3}t^2$. The interval $[c - \delta, c + \delta]$ lies in $[-2, 2]$. Integration of these derivative bounds over the parameter intervals $[c, c + \delta]$ and $[c - \delta, c]$ proves (3.6). □

3.3. **A quantitative comparison lemma.**

Lemma 3.3 (Homogeneous replacement after freezing the source)**.** *Suppose* $g \in C(B_1^n)$ *and* $v \in C(\overline{B_1^n})$ *solves* $-\Delta_\infty v = g$ *in* B_1^n *and*

$$|g - c| \le \delta \quad \text{in } B_1^n, \qquad \delta \ge 0, \qquad |c| + \delta \le 2. \tag{3.7}$$

Define $V(y, t) = v(y) + W_c(t)$ *in* B_1^{n+1}*, and let* h *be the homogeneous Dirichlet solution with boundary values* V*. Then*

$$\|h - V\|_{L^\infty(B_1^{n+1})} \le \delta, \qquad \|h(\,\cdot\,, 0) - v\|_{L^\infty(B_1^n)} \le \delta. \tag{3.8}$$

Proof. For $(y, t) \in \overline{B_1^{n+1}}$, one has $y \in \overline{B_1^n}$ and $|t| \le 1$. Consequently V extends continuously to the closed $(n+1)$-dimensional ball, and the homogeneous Dirichlet solution $h \in C(\overline{B_1^{n+1}})$ exists by (2.2). Define on the same closed ball

$$S(y, t) = v(y) + W_{c+\delta}(t) - \delta, \qquad T(y, t) = v(y) + W_{c-\delta}(t) + \delta.$$

Lemmas 3.1 and 3.2 give, in the viscosity sense,

$$-\Delta_\infty S = g - c - \delta \le 0, \qquad -\Delta_\infty T = g - c + \delta \ge 0.$$

By (3.6), throughout the closed ball,

$$V - \delta \le S \le V \le T \le V + \delta. \tag{3.9}$$

Thus S and T are continuous on the closed ball and respectively a subsolution and a supersolution of $-\Delta_\infty h = 0$ in the open ball. Since $S \le V = h \le T$ on the boundary, (2.3) gives $S \le h \le T$. Combining this with (3.9) proves the first bound in (3.8). The second follows by taking $t = 0$ and using $W_c(0) = 0$. □

Remark 3.4. The error in (3.8) is linear in the variation δ of the source. The comparison is entirely for the homogeneous equation, so it does not require a comparison or uniqueness theorem for a sign-changing inhomogeneous source. The slice $h(\,\cdot\,, 0)$ is used only as a restriction of a regular $(n + 1)$-dimensional function; no equation for that slice is asserted.

4. Affine approximation and proof of the main theorem

4.1. Two-scale affine approximation.

Proposition 4.1. *Let $0 < \sigma \le 1$ and $f \in C^{0,\sigma}(B_1^n)$. Suppose $u \in C(B_1^n)$ solves $-\Delta_\infty u = f$ and*

$$\operatorname{osc}_{B_1} u \le 1, \qquad \|f\|_{L^\infty(B_1)} + [f]_{\sigma;B_1} \le 1. \tag{4.1}$$

There is a constant $C = C(n, a, C_H)$ such that, for every $x_0 \in B_{1/2}$ and $0 < r \le 1/8$, there is a vector $p_{x_0,r}$ with $|p_{x_0,r}| \le C$ satisfying, for every $0 < \rho \le r/2$,

$$\sup_{|z|\le\rho} |u(x_0+z) - u(x_0) - p_{x_0,r}\cdot z| \le C\left(r^{2+\sigma} + \rho^{1+a} r^{-a}\right). \tag{4.2}$$

The estimate holds on the full ball $B_\rho(x_0)$.

If instead $u \in C(B_1^n)$ solves the equation with $f \in C(B_1^n) \cap L^\infty(B_1^n)$ and (2.5) holds, the same conclusion holds with (4.2) replaced by

$$\sup_{|z|\le\rho} |u(x_0+z) - u(x_0) - p_{x_0,r}\cdot z| \le C\left(r^{2} + \rho^{1+a} r^{-a}\right). \tag{4.3}$$

This alternative requires no Hölder assumption on f and has the same constant dependence and uniform slope bound.

Proof. By Lemma 2.3, $\operatorname{Lip}(u; B_{3/4}) \le L$. Define

$$u_r(y) = \frac{u(x_0+ry) - u(x_0)}{r}, \qquad g_r(y) = r f(x_0+ry). \tag{4.4}$$

The closed ball $\overline{B_r(x_0)}$ lies strictly inside $B_{3/4}$, so $u_r \in C(\overline{B_1^n})$, $u_r(0) = 0$, and $\|u_r\|_\infty \le L$. Lemma 2.2, with amplitude factor $A = r$, gives

$$-\Delta_\infty u_r = g_r \qquad \text{in } B_1^n. \tag{4.5}$$

In the Hölder case, set

$$c_r = r f(x_0), \qquad \delta_r = r^{1+\sigma}[f]_{\sigma;B_1}. \tag{4.6}$$

Then $|g_r - c_r| \le \delta_r \le r^{1+\sigma}$ and $|c_r| + \delta_r \le r + r^{1+\sigma} \le 2$. In the bounded-source case, set $c_r = 0$ and $\delta_r = r$; the bound $|g_r| \le r$ gives the same freezing conditions. In either case Lemma 3.3 provides an infinity-harmonic function h_r in B_1^{n+1} with boundary values $u_r(y) + W_{c_r}(t)$ and

$$\|h_r - (u_r + W_{c_r})\|_{L^\infty(B_1^{n+1})} \le \delta_r. \tag{4.7}$$

The oscillation of its boundary values is at most $2L + 2\cdot 14^{1/3}$. By the homogeneous maximum principle and Theorem 1.1,

$$\|Dh_r\|_{L^\infty(B_{1/2}^{n+1})} + [Dh_r]_{a;B_{1/2}^{n+1}} \le C. \tag{4.8}$$

Let $p_{x_0,r} = D_y h_r(0,0)$. For $|y| < 1/2$, the whole segment $\{(sy, 0) : 0 \le s \le 1\}$ lies in $B_{1/2}^{n+1}$. The fundamental theorem of calculus and (4.8) give

$$\begin{aligned} |h_r(y,0) - h_r(0,0) - p_{x_0,r}\cdot y| &\le |y| \int_0^1 |D_y h_r(sy,0) - D_y h_r(0,0)|\, ds \\ &\le C|y|^{1+a} \int_0^1 s^a\, ds \le C|y|^{1+a}. \end{aligned}$$

The estimate at $|y| = 1/2$ follows by continuity from smaller radii. At $t = 0$ all the profiles vanish. Applying (4.7) at both $(y,0)$ and $(0,0)$, we obtain

$$|u_r(y) - p_{x_0,r}\cdot y| \le 2\delta_r + C|y|^{1+a} \qquad (|y| \le 1/2). \tag{4.9}$$

The two comparison errors account for anchoring the affine approximation at $u_r(0) = 0$. Put $y = z/r$ and multiply (4.9) by r. The resulting error is $2r\delta_r + C\rho^{1+a} r^{-a}$. Since $r\delta_r \le r^{2+\sigma}$ in the

Hölder case and $r\delta_r = r^2$ in the bounded-source case, this proves (4.2) and (4.3), respectively. The bound for $p_{x_0,r}$ follows from (4.8). □

4.2. **From uniform affine approximation to a Hölder gradient.** The following elementary form of the affine-approximation criterion is stated with full small balls, so the domain issue in comparing expansions at nearby centers is explicit.

Lemma 4.2. *Let $0 < \gamma < 1$, $0 < \rho_* \le 1/4$, $K, M \ge 0$, and $u \in C(B_1^n)$. Suppose that for every $x \in B_{1/2}$ and $0 < \rho \le \rho_*$ there is a vector $p(x,\rho)$ such that*

$$|p(x,\rho)| \le K, \qquad \sup_{|z|\le\rho} |u(x+z) - u(x) - p(x,\rho)\cdot z| \le M\rho^{1+\gamma}. \tag{4.10}$$

Then u is differentiable at every point of $B_{1/2}$,

$$|u(x+z) - u(x) - Du(x)\cdot z| \le C_\gamma M|z|^{1+\gamma} \quad (x \in B_{1/2},\ |z| \le \rho_*), \tag{4.11}$$

and

$$\|Du\|_{L^\infty(B_{1/2})} \le K, \qquad [Du]_{\gamma;B_{1/2}} \le C_\gamma\left(M + K\rho_*^{-\gamma}\right). \tag{4.12}$$

Proof. For a fixed center x, let $\rho_j = 2^{-j}\rho_*$ and $p_j = p(x,\rho_j)$. Comparing the two affine approximations on $\overline{B_{\rho_{j+1}}(0)}$ gives

$$\rho_{j+1}|p_j - p_{j+1}| \le M\rho_j^{1+\gamma} + M\rho_{j+1}^{1+\gamma}.$$

Therefore

$$|p_j - p_{j+1}| \le 4M\rho_j^\gamma. \tag{4.13}$$

The geometric series converges, so p_j tends to a vector $P(x)$, uniformly in the center, and

$$|P(x) - p_j| \le \frac{4M}{1-2^{-\gamma}}\rho_j^\gamma. \tag{4.14}$$

Because $|p_j| \le K$, also $|P(x)| \le K$. For $0 < |z| \le \rho_*$, choose j with $\rho_{j+1} < |z| \le \rho_j$. Equations (4.10) and (4.14) give

$$\begin{aligned}|u(x+z) - u(x) - P(x)\cdot z| &\le M\rho_j^{1+\gamma} + \frac{4M}{1-2^{-\gamma}}\rho_j^\gamma|z| \\ &\le C_\gamma M|z|^{1+\gamma}.\end{aligned}$$

Thus u is differentiable at x, $Du(x) = P(x)$, and (4.11) holds.

For the Hölder estimate, take distinct $x, y \in B_{1/2}$ and write $d = |x-y|$. If $2d \le \rho_*$, fix a unit vector e and put $z = y + de$. Use (4.11) at y with increment de, and at x with increments $y - x$ and $z - x$. Equivalently, writing $E_x(h) = u(x+h) - u(x) - Du(x)\cdot h$, subtraction gives

$$d\,(Du(y) - Du(x))\cdot e = E_x(y - x + de) - E_x(y - x) - E_y(de).$$

Hence

$$|(Du(y) - Du(x))\cdot e| \le C_\gamma M d^\gamma.$$

The point z need not belong to $B_{1/2}$: the expansions in (4.11) are valid on full balls about their centers, and all increments just used have length at most $2d \le \rho_*$. Moreover, $x + \overline{B_{\rho_*}}$ and $y + \overline{B_{\rho_*}}$ lie in B_1 because $x, y \in B_{1/2}$ and $\rho_* \le 1/4$. Taking the supremum over e proves the desired estimate for small d. For $2d > \rho_*$, the bound $|Du| \le K$ gives

$$\frac{|Du(x) - Du(y)|}{|x-y|^\gamma} \le 2^{1+\gamma}K\rho_*^{-\gamma}.$$

This proves (4.12). □

4.3. **Balancing the scales and restoring the data.**

Proof of Theorem 1.2. First work in B_1. Set

$$A = \operatorname{osc}_{B_1} u + \left(\|f\|_{L^\infty(B_1)} + [f]_{\sigma;B_1} \right)^{1/3}. \tag{4.15}$$

If $A = 0$, u is constant and the conclusion is immediate. Otherwise define

$$v = \frac{u - u(0)}{A}, \qquad \widetilde{f} = \frac{f}{A^3}.$$

Then $-\Delta_\infty v = \widetilde{f}$ and

$$\operatorname{osc}_{B_1} v \le 1, \qquad \|\widetilde{f}\|_{L^\infty(B_1)} + [\widetilde{f}]_{\sigma;B_1} \le 1.$$

We prove a uniform estimate for v using Proposition 4.1.

Let

$$\theta = \frac{1+a}{2+\sigma+a} \in (0,1). \tag{4.16}$$

Choose $0 < \rho_* \le 1/4$, depending only on a, σ, so small that for every $0 < \rho \le \rho_*$ the radius $r = \rho^\theta$ satisfies $r \le 1/8$ and $\rho \le r/2$. For example, one may take

$$\rho_* = \min\left\{ \frac{1}{8}, (1/8)^{1/\theta}, 2^{-1/(1-\theta)} \right\}.$$

The identity

$$\theta(2+\sigma) = 1 + a - a\theta = 1 + \frac{a(1+\sigma)}{2+\sigma+a} = 1 + \beta \tag{4.17}$$

shows that both terms in (4.2) have order $\rho^{1+\beta}$. Thus for every $x \in B_{1/2}$ and $0 < \rho \le \rho_*$ there is a vector $p(x,\rho)$, uniformly bounded, with

$$\sup_{|z| \le \rho} |v(x+z) - v(x) - p(x,\rho)\cdot z| \le C\rho^{1+\beta}.$$

Since $0 < \beta < a \le 1/3$, Lemma 4.2 applies with $\gamma = \beta$ and the uniform slope and error bounds just obtained. It gives

$$\|Dv\|_{L^\infty(B_{1/2})} + [Dv]_{\beta;B_{1/2}} \le C.$$

The constant depends only on n, a, C_H, σ, since β and ρ_* depend only on a, σ. Multiplication by A proves

$$\|Du\|_{L^\infty(B_{1/2})} + [Du]_{\beta;B_{1/2}} \le C\left\{ \operatorname{osc}_{B_1} u + (\|f\|_\infty + [f]_{\sigma;B_1})^{1/3} \right\}. \tag{4.18}$$

For a general ball, put $U(y) = u(x_* + Ry)$ and $F(y) = R^4 f(x_* + Ry)$. Then $-\Delta_\infty U = F$ in B_1, and

$$\|F\|_{L^\infty(B_1)} = R^4 F_R, \qquad [F]_{\sigma;B_1} = R^{4+\sigma} H_{R,\sigma}.$$

The unit-ball result gives $U \in C^{1,\beta}(B_{1/2})$, hence $u \in C^{1,\beta}(B_{R/2}(x_*))$ under the inverse affine change of variables. The chain rule now gives

$$\|DU\|_{L^\infty(B_{1/2})} = R\|Du\|_{L^\infty(B_{R/2}(x_*))}, \qquad [DU]_{\beta;B_{1/2}} = R^{1+\beta}[Du]_{\beta;B_{R/2}(x_*)}.$$

Substituting these identities into (4.18) for U gives (1.6). □

5. Companion estimates and the exponent obstruction

5.1. Continuous bounded sources.

Theorem 5.1. *Let $u \in C(B_R(x_*))$ be a bounded viscosity solution of $-\Delta_\infty u = f$, where $f \in C(B_R(x_*)) \cap L^\infty(B_R(x_*))$. With $\beta_0 = a/(2+a)$, one has $u \in C^{1,\beta_0}(B_{R/2}(x_*))$ and*

$$\begin{aligned} R\|Du\|_{L^\infty(B_{R/2}(x_*))} + R^{1+\beta_0}[Du]_{\beta_0;B_{R/2}(x_*)} \\ \le C\left\{\operatorname{osc}_{B_R(x_*)} u + (R^4\|f\|_{L^\infty(B_R(x_*))})^{1/3}\right\}. \end{aligned} \tag{5.1}$$

The constant depends only on the dimension and the fixed homogeneous theorem.

Proof. On the unit ball put $A_0 = \operatorname{osc}_{B_1} u + \|f\|^{1/3}_{L^\infty(B_1)}$. If $A_0 = 0$, u is constant and the conclusion is immediate. Otherwise replace u by $(u - u(0))/A_0$ and f by f/A_0^3; Lemma 2.2 preserves the equation. Relabeling these normalized functions, we have $\operatorname{osc} u \le 1$ and $\|f\|_\infty \le 1$, so the bounded-source alternative of Proposition 4.1 applies. It gives (4.3), with uniformly bounded slopes, for every $x_0 \in B_{1/2}$, $0 < r \le 1/8$, and $0 < \rho \le r/2$.

Set $\theta_0 = (1+a)/(2+a) \in (0,1)$ and choose $\rho_{*,0}$ by the radius formula following (4.16), with θ replaced by θ_0. This radius depends only on a. For $0 < \rho \le \rho_{*,0}$, the choice $r = \rho^{\theta_0}$ satisfies $r \le 1/8$ and $\rho \le r/2$. Since $2\theta_0 = 1 + a - a\theta_0 = 1 + \beta_0$, both terms in (4.3) have order $\rho^{1+\beta_0}$. Lemma 4.2, with uniform K, M and the chosen radius $\rho_{*,0}$, proves the normalized estimate. Multiplication by A_0 restores the original amplitude. Finally, $U(y) = u(x_* + Ry)$ has source $R^4 f(x_* + Ry)$; the derivative scaling in the proof of Theorem 1.2 gives (5.1). □

Remark 5.2. A continuous viscosity solution u and its continuous source f are bounded on compact subsets. Thus Theorem 5.1 gives local C^{1,β_0} regularity for every continuous source on an open domain. Its constant does not depend on a modulus of continuity of f. Continuity is retained in the formulation of the viscosity equation; no assertion for arbitrary discontinuous sources is needed here.

5.2. A common exponent for the source and the gradient.

Corollary 5.3. *Define $\alpha_* = \sqrt{1+a} - 1$. If $0 < \alpha \le \alpha_*$ and $f \in C^{0,\alpha}_{\mathrm{loc}}(\Omega)$, then every continuous viscosity solution of $-\Delta_\infty u = f$ belongs to $C^{1,\alpha}_{\mathrm{loc}}(\Omega)$. On every ball $B_R(x_*) \Subset \Omega$ with the corresponding finite data norms, the estimate obtained from (1.6) with $\sigma = \alpha$ by lowering the gradient exponent from $\beta(\alpha) = a(1+\alpha)/(2+\alpha+a)$ to α holds, with a constant depending only on n, α and the fixed homogeneous constants.*

Proof. First restrict to any ball $B_R(x_*)$ compactly contained in Ω on which u is bounded and $[f]_{\alpha;B_R(x_*)}$ is finite. Such balls can be centered at every point of Ω, so their concentric half-balls cover Ω. Since $0 < \alpha \le \alpha_* < 1$, apply Theorem 1.2 with $\sigma = \alpha$, and denote its gradient exponent by $\beta(\alpha)$. Then

$$\frac{a(1+\alpha)}{2+\alpha+a} - \alpha = \frac{a - 2\alpha - \alpha^2}{2+\alpha+a} \ge 0.$$

The diameter of $B_{R/2}(x_*)$ is R, so

$$[Du]_{\alpha;B_{R/2}(x_*)} \le R^{\beta(\alpha)-\alpha}[Du]_{\beta(\alpha);B_{R/2}(x_*)}.$$

Consequently $R^{1+\alpha}[Du]_{\alpha;B_{R/2}} \le R^{1+\beta(\alpha)}[Du]_{\beta(\alpha);B_{R/2}}$. Substitution into (1.6) proves the claimed scaled estimate and, by the local covering, the asserted regularity on Ω. □

For example, choosing the source exponent $\sigma = a$ in the main theorem gives exactly $\beta = a/2$. The range in Corollary 5.3 is a sufficient range produced by this proof; it is not asserted to be a sharp threshold.

5.3. **Constant sources and exact lifting.**

Proposition 5.4. *If $-\Delta_\infty u = c$ in $B_R(x_*)$, where c is constant and u is a bounded continuous viscosity solution, then $u \in C^{1,a}(B_{R/2}(x_*))$ and*

$$\begin{aligned} R\|Du\|_{L^\infty(B_{R/2}(x_*))} + R^{1+a}[Du]_{a;B_{R/2}(x_*)} \\ \leq C\left\{\operatorname{osc}_{B_R(x_*)} u + (R^4|c|)^{1/3}\right\}. \end{aligned} \tag{5.2}$$

The constant C depends only on n and the fixed homogeneous estimate.

Proof. In B_1, normalize by $A = \operatorname{osc}_{B_1} u + |c|^{1/3}$, with the zero case immediate, and put $v = (u - u(0))/A$ and $\widetilde{c} = c/A^3$. Then $\operatorname{osc} v \leq 1$, $|\widetilde{c}| \leq 1$, and $-\Delta_\infty v = \widetilde{c}$. By Lemmas 3.1 and 3.2,

$$V(y,t) = v(y) + W_{\widetilde{c}}(t)$$

is exactly infinity-harmonic in B_1^{n+1}. Its oscillation is at most $1 + 2 \cdot 14^{1/3}$. Apply Theorem 1.1 to obtain $V \in C^{1,a}(B_{1/2}^{n+1})$. Since $v(y) = V(y,0)$, differentiability of V along horizontal increments gives differentiability of v and $Dv(y) = D_y V(y,0)$. For $y, z \in B_{1/2}^n$,

$$|Dv(y) - Dv(z)| \leq |DV(y,0) - DV(z,0)| \leq [DV]_{a;B_{1/2}^{n+1}} |y - z|^a.$$

The same projection bounds $|Dv|$ by $|DV|$, so $\|Dv\|_{L^\infty(B_{1/2})} + [Dv]_{a;B_{1/2}} \leq C$. Undoing the amplitude normalization gives the estimate on B_1. For a general ball, apply this estimate to $U(y) = u(x_* + Ry)$, whose constant source is $R^4 c$. The derivative scaling from the proof of Theorem 1.2 then gives (5.2). □

5.4. **The obstruction at exponent** $1/3$**.**

Proposition 5.5. *For every $n \geq 2$, the function*

$$u(x) = x_2 - \frac{3^{4/3}}{4}|x_1|^{4/3} \tag{5.3}$$

is a viscosity solution of $-\Delta_\infty u = 1$ in $\mathbb{R}^n$, satisfies $|Du| \geq 1$, and, for every $1/3 < \eta \leq 1$, does not belong to $C^{1,\eta}$ in any neighborhood of the origin.

Proof. Define $q(t) = -(3^{4/3}/4)|t|^{4/3}$. For $t \neq 0$,

$$q'(t) = -3^{1/3}\operatorname{sgn}(t)|t|^{1/3}, \qquad q''(t) = -3^{-2/3}|t|^{-2/3}, \qquad -(q')^2 q'' = 1.$$

At $t = 0$, the function q is differentiable with $q'(0) = 0$. Every upper C^2 test has derivative zero and satisfies the subsolution inequality $0 \leq 1$. There is no lower C^2 test. Indeed, a touching test ϕ would have $\phi(0) = \phi'(0) = 0$, and Taylor's theorem would give $\phi(t) \geq -Ct^2$ for small t. With $K = 3^{4/3}/4 > 0$, one has $Ct^2 < K|t|^{4/3}$ for all sufficiently small nonzero t, so $\phi(t) \geq -Ct^2 > -K|t|^{4/3} = q(t)$, contradicting contact from below. Thus the supersolution condition at the origin is vacuous, and q solves $-(q')^2 q'' = 1$ in the viscosity sense on $\mathbb{R}$.

Apply Lemma 3.1 with the smooth function $w(t) = t$ to add the second variable. Further unused variables can be added by the same lemma with $w = 0$. This proves the equation for (5.3) in every dimension $n \geq 2$. Direct differentiation gives

$$Du(x) = \left(-3^{1/3}\operatorname{sgn}(x_1)|x_1|^{1/3}, 1, 0, \ldots, 0\right).$$

In particular, $|Du| \geq 1$ and

$$|Du(he_1) - Du(0)| = 3^{1/3}|h|^{1/3}.$$

For $1/3 < \eta \leq 1$, dividing by $|h|^\eta$ and sending $h \to 0$ proves the failure of $C^{1,\eta}$ regularity. □

Remark 5.6 (Scope of the exponent transfer). The main exponent is strictly smaller than the homogeneous exponent: $\beta/a = (1+\sigma)/(2+\sigma+a) < 1$. This loss comes from balancing the freezing error with the homogeneous Taylor error. Proposition 5.4 shows that there is no such loss for constant sources and gives an endpoint $1/3$ estimate whenever the homogeneous input has $a = 1/3$. For variable sources, the present argument neither determines the optimal exponent nor provides an endpoint $1/3$ estimate. It uses the homogeneous theorem in dimension $n+1$: for a forced equation in the plane, the input is a_3, so the planar homogeneous endpoint results do not directly replace that input. Any improvement of the homogeneous estimate in dimension $n+1$ can be inserted into the same transfer formula.

References

[1] G. Aronsson, M. G. Crandall, and P. Juutinen, A tour of the theory of absolutely minimizing functions, *Bull. Amer. Math. Soc. (N.S.)* **41** (2004), no. 4, 439–505. doi:10.1090/S0273-0979-04-01035-3.

[2] S. N. Armstrong and C. K. Smart, An easy proof of Jensen's theorem on the uniqueness of infinity harmonic functions, *Calc. Var. Partial Differential Equations* **37** (2010), 381–384. doi:10.1007/s00526-009-0267-9.

[3] K. K. Brustad, Infinity-harmonic functions in the plane: Regularity by injectivity, preprint, 2026. arXiv:2606.08257.

[4] G. Crasta and I. Fragalà, On the Dirichlet and Serrin problems for the inhomogeneous infinity Laplacian in convex domains: Regularity and geometric results, *Arch. Ration. Mech. Anal.* **218** (2015), 1577–1607. doi:10.1007/s00205-015-0888-4.

[5] L. C. Evans and O. Savin, $C^{1,\alpha}$ regularity for infinity harmonic functions in two dimensions, *Calc. Var. Partial Differential Equations* **32** (2008), no. 3, 325–347. doi:10.1007/s00526-007-0143-4.

[6] L. C. Evans and C. K. Smart, Everywhere differentiability of infinity harmonic functions, *Calc. Var. Partial Differential Equations* **42** (2011), 289–299. doi:10.1007/s00526-010-0388-1.

[7] G. Hong and X. Feng, Superposition principle on the viscosity solutions of infinity Laplace equations, *Nonlinear Anal.* **171** (2018), 32–40. doi:10.1016/j.na.2018.01.011.

[8] R. Jensen, Uniqueness of Lipschitz extensions: Minimizing the sup norm of the gradient, *Arch. Ration. Mech. Anal.* **123** (1993), no. 1, 51–74. doi:10.1007/BF00386368.

[9] H. Koch, Y. R.-Y. Zhang, and Y. Zhou, Some sharp Sobolev regularity for inhomogeneous infinity Laplace equation in plane, *J. Math. Pures Appl. (9)* **132** (2019), 483–521. doi:10.1016/j.matpur.2019.02.007.

[10] E. Lindgren, On the regularity of solutions of the inhomogeneous infinity Laplace equation, *Proc. Amer. Math. Soc.* **142** (2014), no. 1, 277–288. doi:10.1090/S0002-9939-2013-12180-5.

[11] Q. Liu, J. J. Manfredi, and X. Zhou, Superposition property in disjoint variables for the infinity Laplace equation, preprint, 2025. arXiv:2508.17106.

[12] G. Lu and P. Wang, Inhomogeneous infinity Laplace equation, *Adv. Math.* **217** (2008), no. 4, 1838–1868. doi:10.1016/j.aim.2007.11.020.

[13] OpenAI, Uniform Interior $C^{1,\alpha}$ Estimates for Infinity-Harmonic Functions, preprint, October 4, 2026. GitHub manuscript.

[14] O. Savin, C^1 regularity for infinity harmonic functions in two dimensions, *Arch. Ration. Mech. Anal.* **176** (2005), no. 3, 351–361. doi:10.1007/s00205-005-0355-8.

[15] A. Sobral, E. V. Teixeira, and J. M. Urbano, Qualitative-to-quantitative regularity for degenerate equations, preprint, 2026. arXiv:2610.13137.

[16] K. Xu, Infinity-harmonic functions and inverse mean curvature flow clusters, preprint, 2026. arXiv:2607.06698.